

Motivation

Consider a Gaussian Process regression setup:

$$y \mid f = f(x) + \mathcal{N}(0, \sigma^2)$$

or $y \mid f \sim \text{Bern}(\sigma(f(x)))$ etc.

$$f(X) \sim \mathcal{GP}(0, K_\theta).$$

This setup is common in:

1. Spatial statistics, meteorology
2. Astrophysics
3. Neuroscience (e.g. time series of firing rates)
4. Spatial transcriptomics

